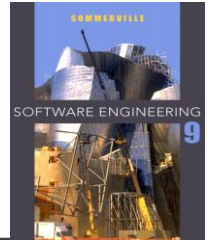


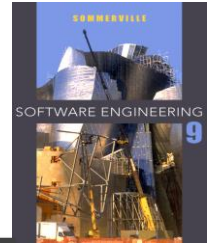
# Grading

# Grading



| Item                                  | Deliverable                        | Grade Percentage |  |
|---------------------------------------|------------------------------------|------------------|--|
| <b>LABORATORY</b>                     |                                    | <b>10.0%</b>     |  |
| <b>ITERATION #1; UX DEMONSTRATION</b> |                                    | <b>5.0%</b>      |  |
| <b>MIDTERM</b>                        |                                    | <b>20.0%</b>     |  |
| <b>WORK PACKAGE 1</b>                 |                                    | <b>10.0%</b>     |  |
| <u>Individual Deliverables</u>        |                                    | 5.0%             |  |
|                                       | User Interface Design (Wireframes) | 2.0%             |  |
|                                       | System Architecture                | 1.5%             |  |
|                                       | UX Architecture                    | 1.0%             |  |
|                                       | Risk Assessment                    | 0.5%             |  |
| <u>Group Deliverables</u>             |                                    | 5.0%             |  |
|                                       | System SRS                         | 2.5%             |  |
|                                       | Project Schedule                   | 0.5%             |  |
|                                       | Personas                           | 1.0%             |  |
|                                       | Coding Standard                    | 0.5%             |  |
|                                       | Defect Tracking                    | 0.5%             |  |

# Laboratory = 10%



| LABORATORY |                                                       |       |        |         |
|------------|-------------------------------------------------------|-------|--------|---------|
| #          | Section Title                                         | Grade | Points | 0.00043 |
|            |                                                       |       |        | 230     |
| 1          | Lab #1: SRS Template                                  | 0.4%  | 10     |         |
| 1          | Lab #1: SDD Template                                  | 0.4%  | 10     |         |
| 1          | Lab #1: Test Plan Template                            | 0.4%  | 10     |         |
| 1          | Lab #1: Software Configuration Management Document    | 0.4%  | 10     |         |
| 2          | Lab #2: Schedule                                      | 0.4%  | 10     |         |
| 2          | Lab #2: Architecture Diagram                          | 0.4%  | 10     |         |
| 2          | Lab #2: Risk Assessment                               | 0.4%  | 10     |         |
| 3          | Lab #3: One Persona                                   | 0.4%  | 10     |         |
| 3          | Lab #3: Interaction Architecture                      | 0.4%  | 10     |         |
| 3          | Lab #3: One Wireframe                                 | 0.4%  | 10     |         |
| 4          | Lab #4: Scrum(Vivify Project)                         | 0.9%  | 20     |         |
| 5          | Lab #5: One User Story per Actor                      | 0.9%  | 20     |         |
| 5          | Lab #5: Coding Standard                               | 0.4%  | 10     |         |
| 6          | Lab #6: One Deployment Diagram                        | 0.4%  | 10     |         |
| 6          | Lab #6: One Class Diagram for any module              | 0.4%  | 10     |         |
| 7          | Lab #7: Class Diagram for each module                 | 0.4%  | 10     |         |
| 7          | Lab #7: One Sequence Diagram for each module          | 0.4%  | 10     |         |
| 8          | Lab #8: Junit test code written for Telephone project | 0.4%  | 10     |         |
| 9          | Lab #9: Test Plan                                     | 0.4%  | 10     |         |
| 10         | Lab #10 - Midterm Review                              | 0.0%  | 0      |         |
| 11         | Lab #11 - Midterm Exam                                | 0.0%  | 0      |         |
| 12         | Lab #12 - Costing                                     | 0.4%  | 10     |         |
| 13         | Lab #13 - Help                                        | 0.4%  | 10     |         |
| 14         | Lab #14 - Work on Project                             | 0.0%  | 0      |         |
| 10.0%      |                                                       | TOTAL | 10.0%  | 230 230 |

Chapter 7 Design and implementation

# UX Demonstration – 5%

---

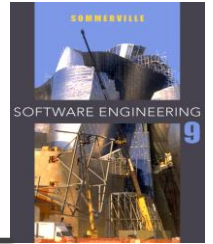


## ✧ Present

- Personas - brief
- Information Architecture
- Interaction Architecture
- Screens

# Work Package 1 – 10%

---



## ✧ **Work Package 1** includes:

- SRS,
- System Architecture,
- Personas,
- Information and Interaction Architectures,
- UI design,
- Coding Standard,
- Defect Tracking Policy & Workflow,
- Schedule with resource assignments
- Risk Assessment.

✧ This is due at the beginning of Lecture 5.

# Take Credit for Your Work

---



## ✧ Printed Submittals

- Group Name
- Student Name
- Title of Document
- If Digital Submission:

File Format:

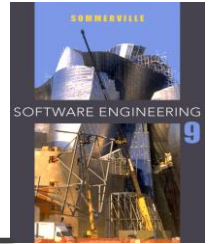
*Group Name-Last Name-First Name-XXX.yyy*

*XXX = SubSRS, Arch, TP*

*XXX = SSRS, Sched, SArch, ATP*

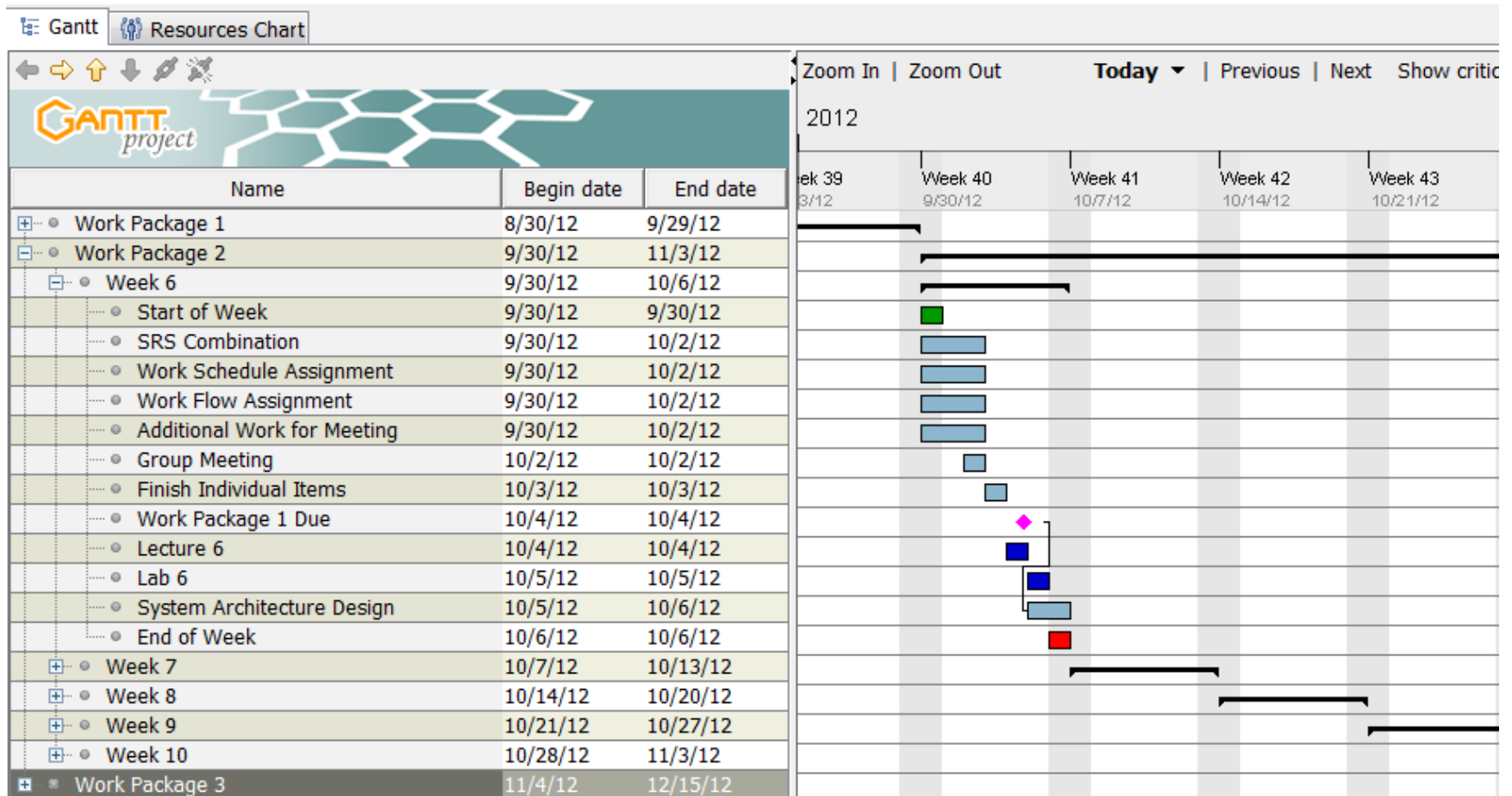
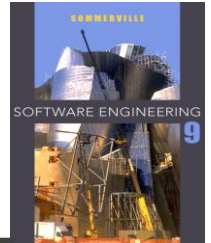
# SRS

---

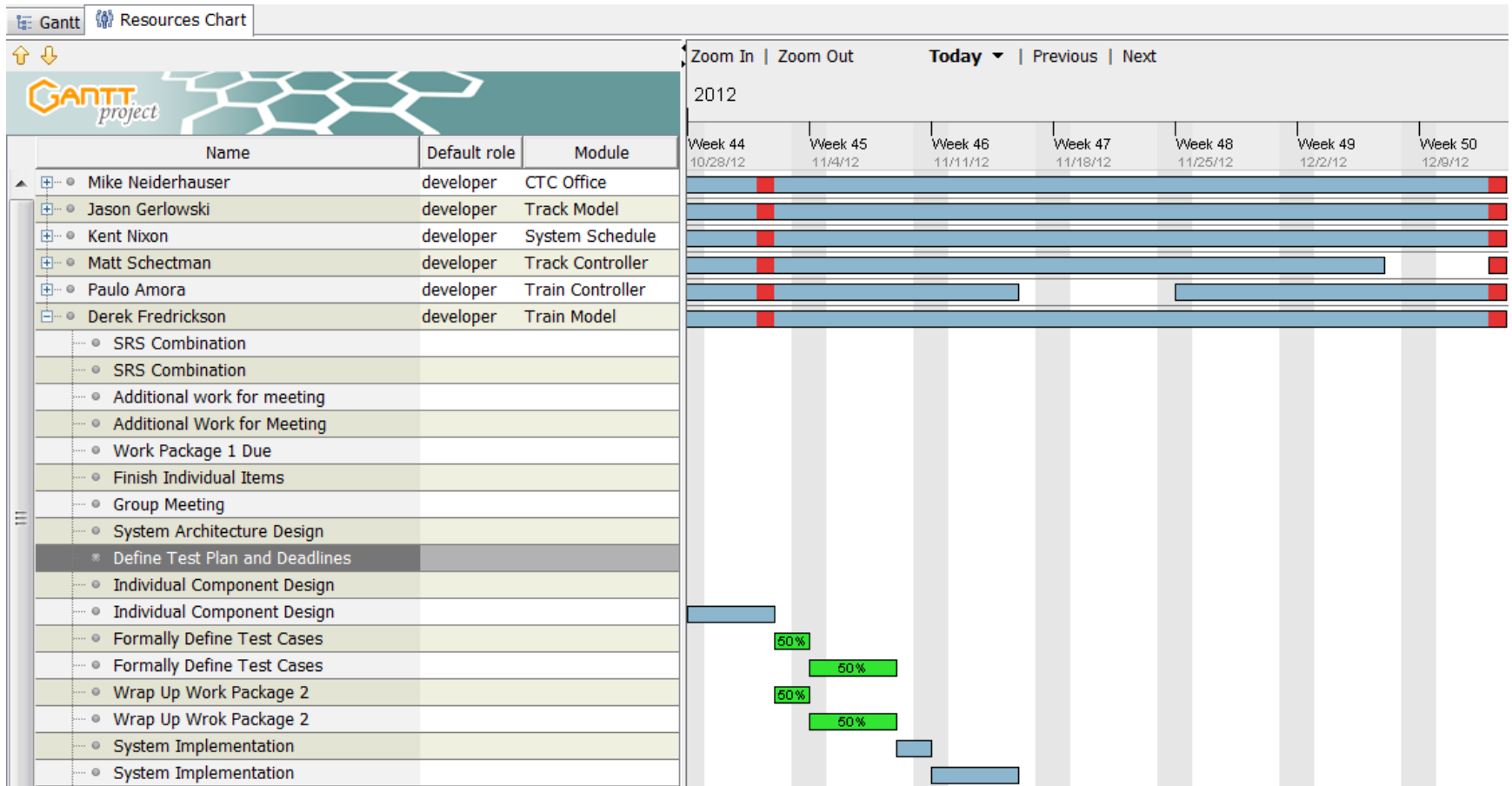
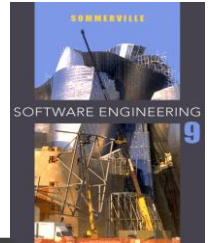


✧ Follow IEEE 830

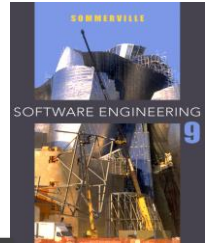
# Schedule



# Schedule



# Coding Standards



## Code Conventions

### 1. File Names

#### 1.1 File Names

Source.cs - C# source file

### 2. Commenting

#### 2.1. Beginning Comments

Comment block should appear at top of class file containing author, revision date, and description.

#### 2.2. Method comments

Comment block should appear above method with name, description of arguments and return variables, and description of function.

#### 2.3. General Code Comments

Line comments should precede control structures, each level of indentation, and large blocks of code. Trailing comments may only be used after individual statements.

### 3. Indentation and White Space

#### 3.1. Code Blocks

Any new code block indicated by an opening curly brace must be indented by one additional tab.

#### 3.2. White Space

Methods should be separated by one line of space. Variable declarations should be separated from other code by one line of space. Related variables should be grouped together. Code for different task should be separated by one line of space.

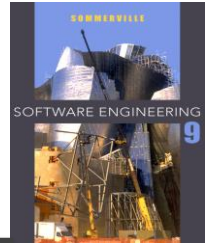
### 4. Names and Declarations

#### 4.1. Variable Declaration

Variables should be declared at the beginning of blocks excluding GUI components. Multiple variables may be declared per line, but variables requiring explanation should be separate.

#### 4.2. Variable Naming

# Defect Tracking Policy



✧ *Numbering Policy*

✧ *Tracking Policy*

✧ *Resolution Policy*

✧ *Tool Used*

## Preliminaries

- Remote code repository hosted by Atlassian's bitbucket © (<https://bitbucket.org/skienite/coe1186-train-project>)
- Bugs will be reported/updated/tracked/resolved via "Issue Tracking" as provided within the bitbucket coe1186-train-project's online interface.
- Each development team member will install Git or Mercurial on their designated machine where they will be developing their code
- Each development team member will create a Bitbucket account.
- The bitbucket repository is listed in the bitbucket dashboard, link → COE1186 Train Project
- We'll use an excel template to enhance the categorization and prioritization of each defect, the "Defect Reporting Template" explains the various definitions.
- The reporting mechanisms and protocols will be subject to change as we enter the integration phase of our coding efforts. For now, we will limit bug reporting to each individual's work on their respective module. Later as team efforts to develop interfaces between modules and such arise, we will enhance the reporting / categorization / assignment / ownership of system defects, reproducing issues, etc.
- This document is subject to changes/enhancements as deemed necessary by the group

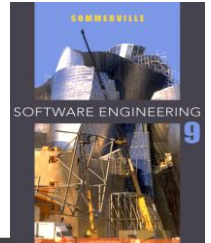
## Reporting / Validating New Coding Defects

1. Open COE1186 Train Project and select the "issues" tab, click on "Create Issue"
2. Input Title (module name) + test case id (when applicable) + a brief description that is more granular in nature and concise.
3. Assign to yourself until we develop integration reporting mechanisms, unless this is a failure within the testing plan specifically. We will assign to another group member who will be responsible for reproducing the error.
4. Priority will be based on the following definitions, which are included in the excel template as well:
  - **Trivial:** A defect, which will not cause an observable product failure or departure from requirement, but has been detected and worthy of noting
  - **Major:** A defect, which will cause and observable product failure or departure from requirements
  - **Minor:** A defect that will not cause failure in execution of the product
  - **Critical:** A defect that will cause the system to crash or close abruptly or effect other software components**NOTE:** We will not be using the priority "Blocker" available through bitbucket's

### Omicron Defect Reporting Template

Name of Reporter: Rob Fitzgerald  
Email Id of Reporter: robertwfitzgeraldjr@gmail.com  
Module or component: Wayside Controller  
Type: coding error  
Priority: 1  
Severity: major  
Work Product: SC - Source Code  
Type of error: Inadequate or sub optimal Design  
Test case id:

# User Interface



Terminal-Velocity (Running) - Microsoft Visual Studio

Terminal Velocity - Track Controller Red

Track Controller Information

Track Controller: 0  
Blocks: 15  
Trains: 0

Prev TC Next TC

Select Track Controller

OK

7 Track Controllers

Terminal Velocity - Track Model

Load File Terminal Velocity: Track Model

Block Info

Block ID: NoBlockSelected  
State: NoBlockSelected  
Heater: NoBlockSelected  
Circuit: NoBlockSelected  
Switch: NoBlockSelected  
Tunnel: NoBlockSelected  
Line: NoBlockSelected

Train ID Route Block

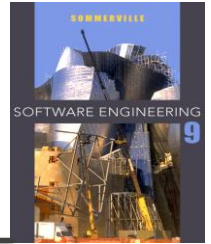
| Train ID | Route Block |
|----------|-------------|
| *        |             |

Block ID State

| Block ID | State  |
|----------|--------|
| 1        | Health |
| 2        | Health |
| 3        | Health |
| 4        | Health |
| 5        | Health |
| 6        | Health |
| 7        | Health |

A pixelated track diagram showing a green loop and a red path. The green loop is a large, irregular shape, and the red path is a smaller, more complex shape that intersects the green loop.

# User Interface



Terminal Velocity - Track Controller Green

Track Controller Information

Track Controller: 0  
Blocks: 12  
Trains: 0

Prev TC Next TC OK

Select Track Controller

8 Track Controllers

|   | Train ID | Route Blocks | Authority | Speed |
|---|----------|--------------|-----------|-------|
| * |          |              |           |       |

|   | Block ID | State   | Switch Destination |
|---|----------|---------|--------------------|
| ▶ | 1        | Healthy | 2                  |
|   | 2        | Healthy | 3                  |
|   | 3        | Healthy | 4                  |
|   | 4        | Healthy | 5                  |
|   | 5        | Healthy | 6                  |
|   | 6        | Healthy | 7                  |
|   | 7        | Healthy | 8                  |

Terminal Velocity - Track Controller Red

Track Controller Information

Track Controller: 0  
Blocks: 15  
Trains: 0

Prev TC Next TC OK

Select Track Controller

7 Track Controllers

|   | Train ID | Route Blocks | Authority | Speed |
|---|----------|--------------|-----------|-------|
| * |          |              |           |       |

|   | Block ID | State   | Switch Destination |
|---|----------|---------|--------------------|
| ▶ | 1        | Healthy | 2                  |
|   | 2        | Healthy | 3                  |
|   | 3        | Healthy | 4                  |
|   | 4        | Healthy | 5                  |
|   | 5        | Healthy | 6                  |
|   | 6        | Healthy | 7                  |
|   | 7        | Healthy | 8                  |

# Risk Assessment & Risk Mitigation Plan

## Risk assessment and mitigation plan

| Risk identification & qualification                                                                                                                                                                                                                                                                                                  | Mitigation plan                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><i>Schedule</i></p> <p>It is possible that a task in the Work Schedule is not completed in due time.<br/>This may happen when a team member lacks the knowledge or skills for the member's duty or is unable to work due to reasons.<br/>The impact is that deadlines are not met, that or some tasks overlap or are delayed.</p> | <p>When setting the Work Schedule, allot 25% more time to each task than expected.<br/>Should that still not be enough time, team members work more hours.</p> |
| <p><i>People</i></p> <p>It is possible that miscommunication between team members or between team and customer occurs.<br/>The impact is that a team member makes the wrong thing.</p>                                                                                                                                               | <p>Set up a discussion board for all team members and the customer.<br/>Team members develop incrementally and contact each other and the customer often.</p>  |
| <p><i>Tools</i></p> <p>It is possible that the code repository shut down temporarily or indefinitely.<br/>The impact is that team members cannot access the code and continue development.</p>                                                                                                                                       | <p>Have a backup of the project in each team member's local computer memory.</p>                                                                               |

| Risk quantification | Negligible   | Marginal        | Critical      |
|---------------------|--------------|-----------------|---------------|
| Likely              | <i>Tools</i> | <i>Schedule</i> |               |
| Unlikely            |              |                 | <i>People</i> |

# Work Package Submittal

---



- ✧ Hand in printed version of everything
  - Except your code in WP3
  - Be sure to put your name on it
- ✧ An electronic Zip file for the group turned on a USB stick
  - I will pass a USB stick around the night work is due

# Questions

---

